EMERGENCY MEDICINE ROTATION REVIEW

CHAPTER 6: ENOT & OPHTHALMOLOGY (HIGH-YIELD GUIDE)

1. ACUTE LARYNGITIS

PATHOPHYSIOLOGY & ETIOLOGY:

Inflammation of the laryngeal mucosa and vocal cords. Most commonly viral (>90% of cases, including Rhinovirus, Influenza, Parainfluenza, and Adenovirus) or triggered by acute vocal overuse/abuse.

CLINICAL PRESENTATION:

Abrupt onset of voice hoarseness (hallmark feature), a dry tickling cough, throat raw pain, and mild odynophagia. Systemic symptoms are usually mild unless part of an influenza prodrome.

PHYSICAL EXAMINATION:

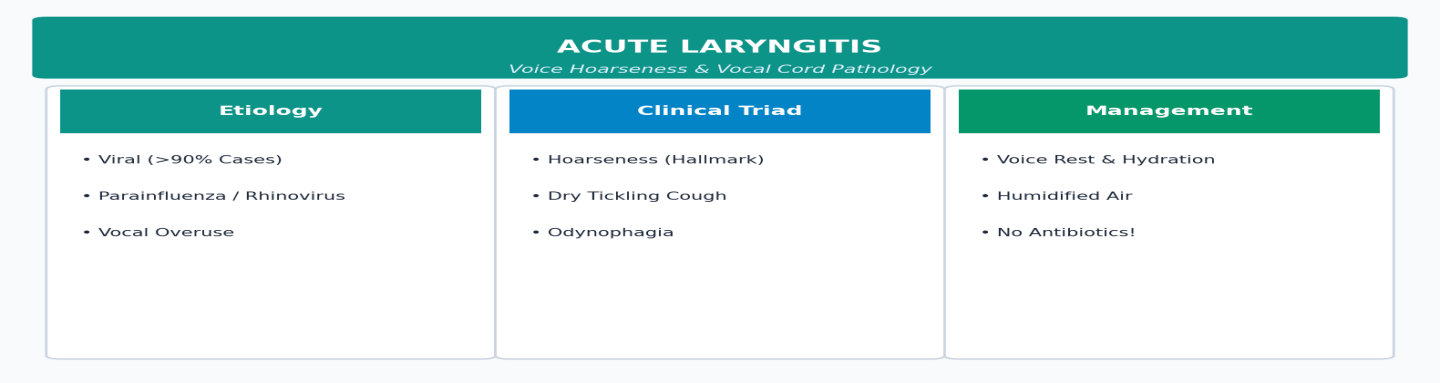
Oropharyngeal examination is typically unremarkable. Indirect laryngoscopy (if performed) reveals diffuse vocal cord erythema, edema, and vascular engorgement with impaired vocal cord closure.

DIAGNOSTIC STUDIES:

Clinical diagnosis. Laryngoscopy is generally NOT indicated unless hoarseness persists for >3 weeks (to rule out vocal cord nodules, polyps, or squamous cell carcinoma).

MANAGEMENT & TREATMENT:

Supportive care is paramount: strict voice rest (avoid whispering as it strains cords), adequate oral hydration, and humidified air. Antibiotics are strictly NOT indicated for viral laryngitis.



BOARD PEARL: Whispering causes greater vocal cord tension and strain than normal quiet speaking. Instruct patients on COMPLETE voice rest!

PATHOPHYSIOLOGY & ETIOLOGY:

Infection of the middle ear space following Eustachian tube dysfunction (often precipitated by a viral URI). Most common bacterial pathogens: *Streptococcus pneumoniae*, nontypeable *Haemophilus influenzae*, and *Moraxella catarrhalis*.

CLINICAL PRESENTATION:

Otalgia (ear pain), ear pulling in infants, fever, irritability, decreased hearing, and occasional otorrhea if tympanic membrane perforation occurs.

PHYSICAL EXAMINATION:

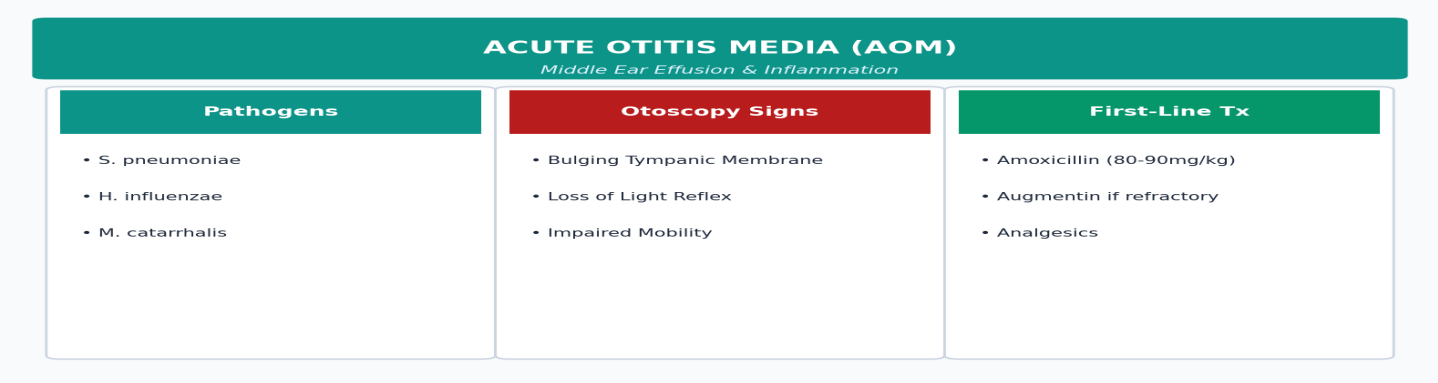
Pneumatic otoscopy reveals a bulging, erythematous tympanic membrane (TM) with loss of normal bony landmarks and light reflex, plus distinctly impaired TM mobility.

DIAGNOSTIC STUDIES:

Clinical diagnosis based on acute onset, middle ear effusion (bulging TM / reduced mobility), and signs of middle ear inflammation (erythema / otalgia).

MANAGEMENT & TREATMENT:

First-line antibiotic: High-dose Amoxicillin (80–90 mg/kg/day). If recent amoxicillin use (<30 days) or concurrent conjunctivitis, prescribe Amoxicillin-Clavulanate (Augmentin). Analgesics (Ibuprofen/Acetaminophen) for pain.



BOARD PEARL: Tympanic membrane BULGING is the most specific physical exam finding for distinguishing AOM from otitis media with effusion (OME).

PATHOPHYSIOLOGY & ETIOLOGY:

Inflammation of the pharynx and tonsils. Most cases are viral (Adenovirus, Enterovirus, EBV). Group A Beta-Hemolytic Streptococcus (GAS / *S. pyogenes*) accounts for 15-30% of pediatric and 5-15% of adult cases.

CLINICAL PRESENTATION:

Sore throat, odynophagia, fever, headache, and anterior cervical lymphadenopathy. Absence of cough favors GAS etiology.

PHYSICAL EXAMINATION:

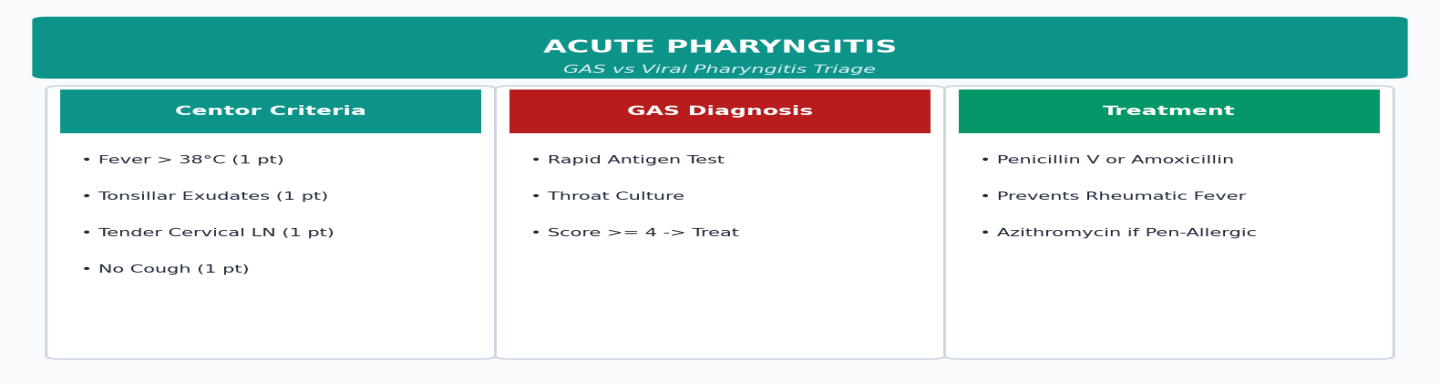
Erythematous pharynx with tonsillar enlargement, exudates, petechiae on the soft palate, and tender anterior cervical lymph nodes. Scarlatiniform rash suggests Scarlet Fever.

DIAGNOSTIC STUDIES:

Apply Modified Centor Criteria (Fever >38°C, Tonsillar Exudates, Tender Ant. Cervical LN, Absence of Cough; +1 pt for age 3-14). Rapid Antigen Detection Test (RADT) or throat culture for score >=2.

MANAGEMENT & TREATMENT:

GAS Treatment: First-line Oral Penicillin V or Amoxicillin for 10 days to prevent Acute Rheumatic Fever. Cephalosporins or Azithromycin for penicillin-allergic patients.



BOARD PEARL: Antibiotic therapy for GAS pharyngitis prevents Acute Rheumatic Fever even if started up to 9 days after symptom onset, but does NOT prevent Post-Streptococcal Glomerulonephritis!

PATHOPHYSIOLOGY & ETIOLOGY:

Inflammation of the paranasal sinuses and nasal cavity lasting <4 weeks. Over 98% are viral (AVRS). Secondary bacterial infection (ABRS) is caused by *S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*.

CLINICAL PRESENTATION:

Nasal congestion, purulent nasal discharge, facial pain/pressure over Maxillary (most common) or Frontal sinuses that worsens when bending forward, and fever.

PHYSICAL EXAMINATION:

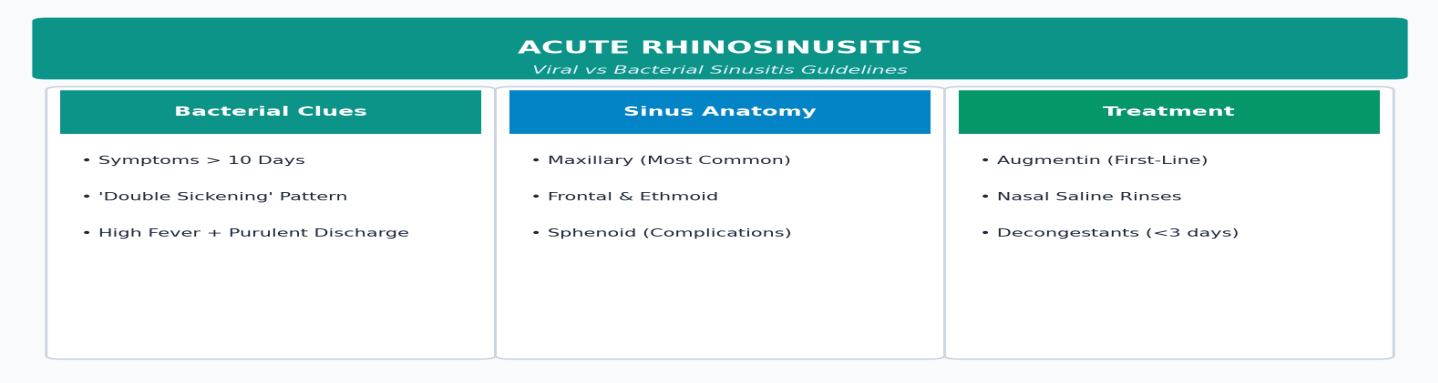
Purulent drainage in the middle meatus, mucosal edema, facial tenderness to percussion over involved sinuses, and transillumination deficits.

DIAGNOSTIC STUDIES:

Clinical diagnosis. ABRS suspected if: (1) Symptoms persist >10 days without improvement; (2) 'Double sickening' (worsening after initial improvement); or (3) Severe onset (Fever >39°C + purulent discharge for 3-4 days).

MANAGEMENT & TREATMENT:

First-line ABRS antibiotic: Amoxicillin-Clavulanate (Augmentin). Doxycycline or Respiratory Fluoroquinolones for PCN allergy. Supportive: Nasal saline irrigation and intranasal steroids.



BOARD PEARL: Routine CT imaging is NOT recommended for uncomplicated rhinosinusitis. Reserve CT for suspected intracranial or orbital complications!

PATHOPHYSIOLOGY & ETIOLOGY:

IgE-mediated Type I hypersensitivity response of the nasal mucosa to environmental allergens (pollen, dust mites, animal dander), causing mast cell degranulation and histamine/leukotriene release.

CLINICAL PRESENTATION:

Paroxysmal sneezing, clear rhinorrhea, nasal congestion, itching of the nose/eyes/palate, and watery red eyes (allergic conjunctivitis).

PHYSICAL EXAMINATION:

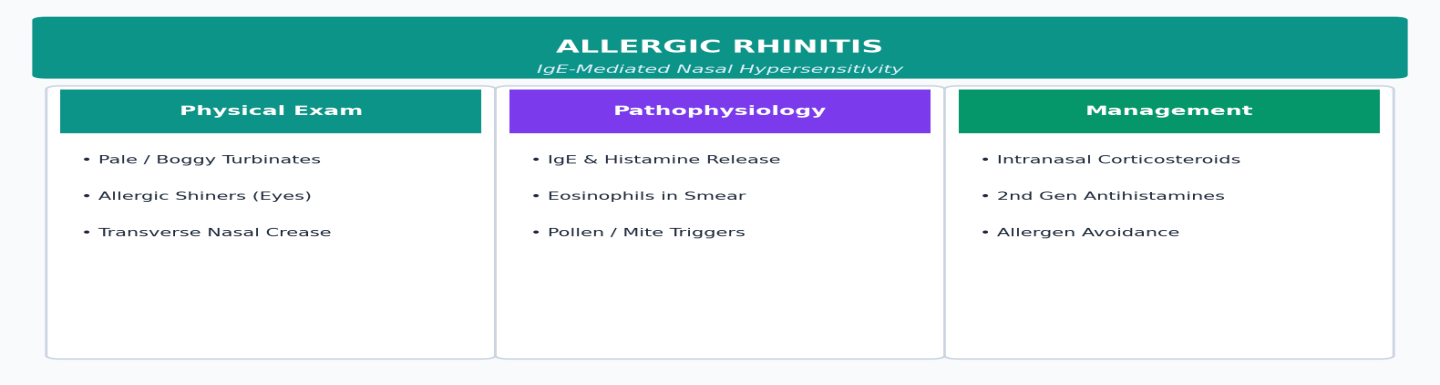
Pale, boggy, bluish nasal turbinates with clear secretions. External signs: 'Allergic shiners' (infraorbital venous congestion), 'Allergic salute' / transverse nasal crease, and Dennie-Morgan lines.

DIAGNOSTIC STUDIES:

Clinical diagnosis based on characteristic history and physical exam. Specific IgE testing (skin prick or RAST) indicated for refractory cases or immunotherapy planning.

MANAGEMENT & TREATMENT:

First-line therapy: Intranasal Corticosteroids (e.g., Fluticasone). Second-generation oral antihistamines (Cetirizine, Loratadine) and nasal saline rinses. Allergen avoidance.



BOARD PEARL: Intranasal corticosteroids are the single most effective monotherapy for allergic rhinitis, outperforming oral antihistamines for nasal congestion!

PATHOPHYSIOLOGY & ETIOLOGY:

Tissue injury of the tympanic membrane (TM) and middle ear resulting from failure of the Eustachian tube to equalize pressure differentials during ambient atmospheric pressure changes (flying, scuba diving).

CLINICAL PRESENTATION:

Sudden onset ear pressure, ear fullness, otalgia, conductive hearing loss, tinnitus, and occasional vertigo or otorrhea if TM ruptures.

PHYSICAL EXAMINATION:

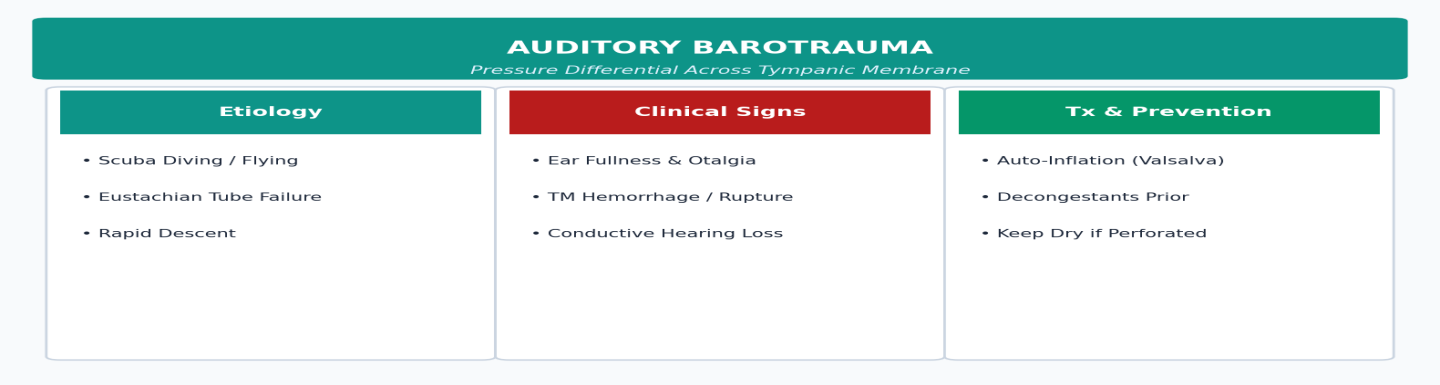
Otoscopy demonstrates TM retraction, mucosal erythema, hemotympanum (blood behind intact TM), or acute TM perforation with bloody fluid in the external canal.

DIAGNOSTIC STUDIES:

Clinical diagnosis based on timing relative to rapid barometric pressure shifts.

MANAGEMENT & TREATMENT:

Prevention/Mild: Equalization maneuvers (Valsalva, swallowing, chewing gum), oral/nasal decongestants prior to flight. Severe/Rupture: Keep ear dry, topical analgesics; most TM perforations heal spontaneously.



BOARD PEARL: Scuba divers with barotrauma who present with inner ear symptoms (vertigo, sensorineural hearing loss) must be evaluated for inner ear barotrauma or decompression sickness!

PATHOPHYSIOLOGY & ETIOLOGY:

Chronic inflammatory condition of the eyelid margins. Anterior blepharitis involves staphylococcal infection or seborrheic dermatitis. Posterior blepharitis involves Meibomian gland dysfunction (strongly associated with Rosacea).

CLINICAL PRESENTATION:

Bilateral eyelid itching, burning, gritty foreign body sensation, morning crusting/matting of eyelids, and red swollen eyelid margins.

PHYSICAL EXAMINATION:

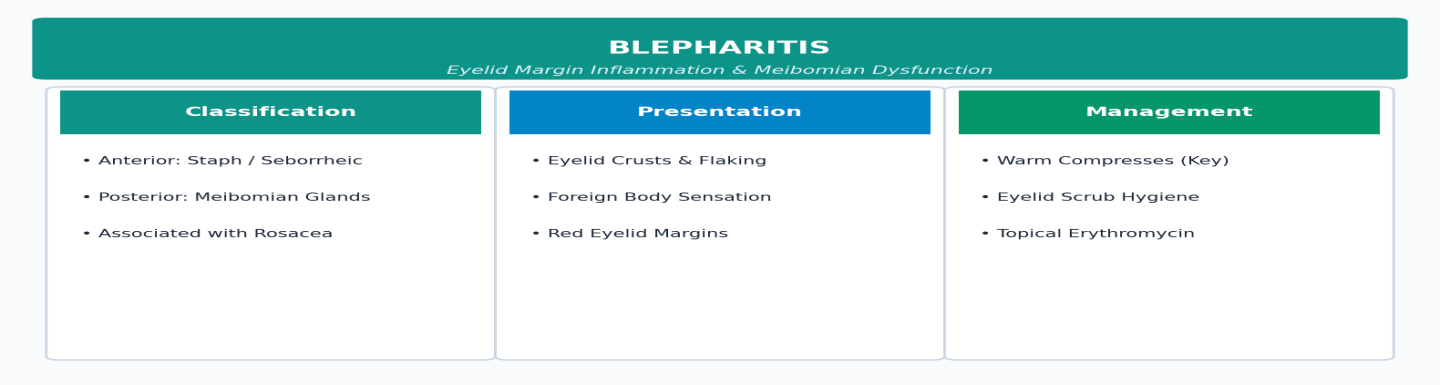
Anterior: Flaking, greasy scales/crusts at eyelash bases, madarosis (lashed loss). Posterior: Hyperemic Meibomian orifice plugging, telangiectasia of lid margins, and altered tear film.

DIAGNOSTIC STUDIES:

Clinical diagnosis via slit-lamp examination.

MANAGEMENT & TREATMENT:

Eyelid hygiene is the cornerstone: Daily warm compresses followed by eyelid margin scrubs with mild baby shampoo. Topical Erythromycin or Bacitracin ointment for anterior; oral Doxycycline for refractory posterior blepharitis.



BOARD PEARL: Chronic posterior blepharitis is a primary cause of recurrent chalazia and hordeola due to obstructed Meibomian gland secretions.

PATHOPHYSIOLOGY & ETIOLOGY:

Fracture of the thin orbital walls (most commonly the inferior orbital floor into the maxillary sinus, or medial orbital wall into ethmoid) due to direct blunt trauma to the orbit.

CLINICAL PRESENTATION:

Periorbital pain, facial trauma, diplopia (especially on upward gaze due to inferior rectus muscle entrapment), enophthalmos, and infraorbital anesthesia (numbness of cheek/upper lip via V2 nerve injury).

PHYSICAL EXAMINATION:

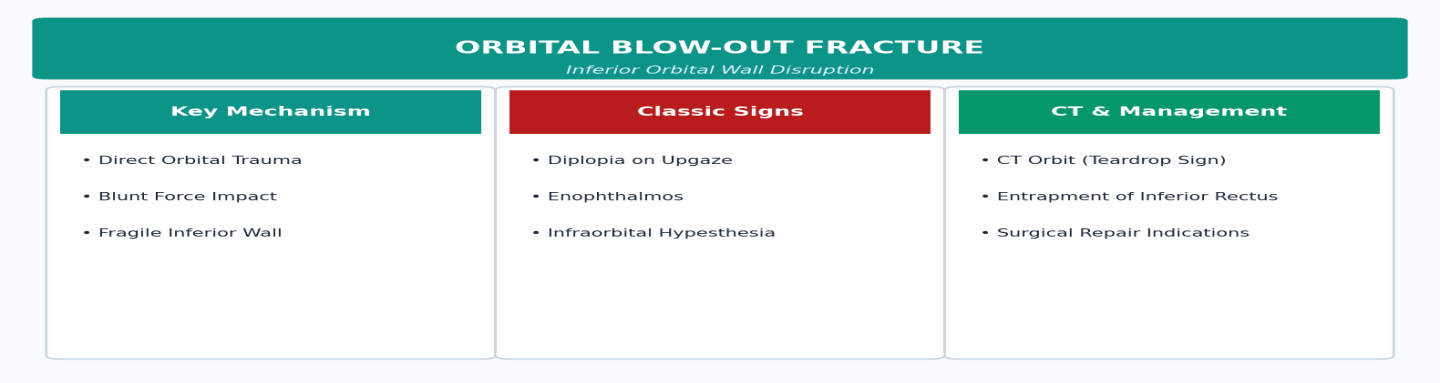
Periorbital ecchymosis, edema, subconjunctival hemorrhage, restricted vertical gaze, and subcutaneous emphysema (worsened by nose blowing).

DIAGNOSTIC STUDIES:

Non-contrast CT Scan of Orbits: Demonstrates orbital floor fracture, 'teardrop sign' (herniated orbital tissue/fat into maxillary sinus), and air-fluid levels.

MANAGEMENT & TREATMENT:

Nasal decongestants, broad-spectrum oral antibiotics (Augmentin), instruct NOT to blow nose. Emergent surgical repair if muscle entrapment causes non-resolving oculocardiac reflex or persistent diplopia.



BOARD PEARL: Always test sensation over the cheek/upper lip — infraorbital nerve (V2) hypesthesia strongly points to an orbital floor fracture!

PATHOPHYSIOLOGY & ETIOLOGY:

Inflammation of the conjunctiva. Bacterial (*S. aureus*, *S. pneumoniae*, *H. influenzae*, *Pseudomonas* in contact lens users). Viral (Adenovirus, highly contagious). Allergic (IgE-mediated).

CLINICAL PRESENTATION:

Red eye, discharge, irritation. Bacterial: Thick purulent/mucopurulent discharge, eyelashes matted shut on awakening. Viral: Watery/serous discharge, burning. Allergic: BILATERAL intense itching.

PHYSICAL EXAMINATION:

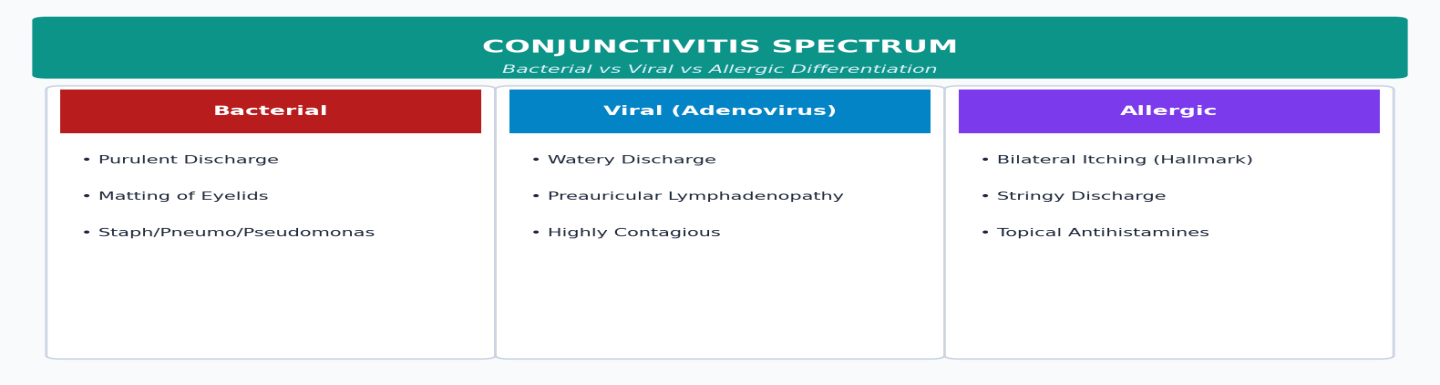
Bacterial: Purulent exudate in conjunctival sac. Viral: Follicular conjunctivitis, ipsilateral tender preauricular lymphadenopathy. Allergic: Cobblestone papillae under palpebral conjunctiva.

DIAGNOSTIC STUDIES:

Clinical diagnosis. Fluorescein staining to rule out corneal abrasion/ulcer.

MANAGEMENT & TREATMENT:

Bacterial: Topical antibiotics (Erythromycin ointment, Polymyxin B/Trimethoprim; Fluoroquinolone like Ciprofloxacin for contact lens wearers). Viral: Warm/cold compresses, artificial tears. Allergic: Topical antihistamines/mast cell stabilizers (Olopatadine).



BOARD PEARL: Contact lens wearers with bacterial conjunctivitis MUST be covered for *Pseudomonas aeruginosa* using topical Fluoroquinolones (e.g., Ciprofloxacin/Ofloxacin)!

PATHOPHYSIOLOGY & ETIOLOGY:

Corneal Abrasion: Loss of corneal epithelial layer due to mechanical trauma or foreign body. Corneal Ulcer (Keratitis): Deeper infection of corneal stroma (commonly *Pseudomonas* in contact lens wearers, or HSV).

CLINICAL PRESENTATION:

Severe sharp eye pain, photophobia, tearing, foreign body sensation, blepharospasm, and decreased visual acuity (if central lesion).

PHYSICAL EXAMINATION:

Fluorescein Staining under Wood's Lamp / Cobalt Blue light reveals yellow-green corneal epithelial defect. Corneal Ulcer shows a focal white/gray opacity (stromal infiltrate). Evert eyelids to search for foreign bodies.

DIAGNOSTIC STUDIES:

Clinical exam with Fluorescein staining. Seidel Test (fluorescein stream under blue light) must be negative to rule out globe perforation.

MANAGEMENT & TREATMENT:

Abrasion: Topical antibiotic drops (Erythromycin or Ciprofloxacin if contact lens wearer), oral NSAIDs for analgesia. Do NOT patch the eye! Corneal Ulcer: Emergent ophthalmology evaluation, intensive topical Fluoroquinolones.



BOARD PEARL: NEVER prescribe topical anesthetic drops (Tetracaine) for home use — they inhibit corneal healing and can lead to corneal melting and blindness!

PATHOPHYSIOLOGY & ETIOLOGY:

Dacryoadenitis: Inflammation/infection of the LACRIMAL GLAND (superolateral orbit; Mumps, EBV, Staph, Sarcoidosis). Dacryocystitis: Infection of the LACRIMAL SAC (inframedial canthus; nasolacrimal duct obstruction; *S. aureus*, *S. pneumo*).

CLINICAL PRESENTATION:

Dacryoadenitis: Superolateral eyelid pain, swelling, tenderness. Dacryocystitis: Sudden pain, redness, and swelling at the medial canthus / inframedial lower eyelid with tearing (epiphora).

PHYSICAL EXAMINATION:

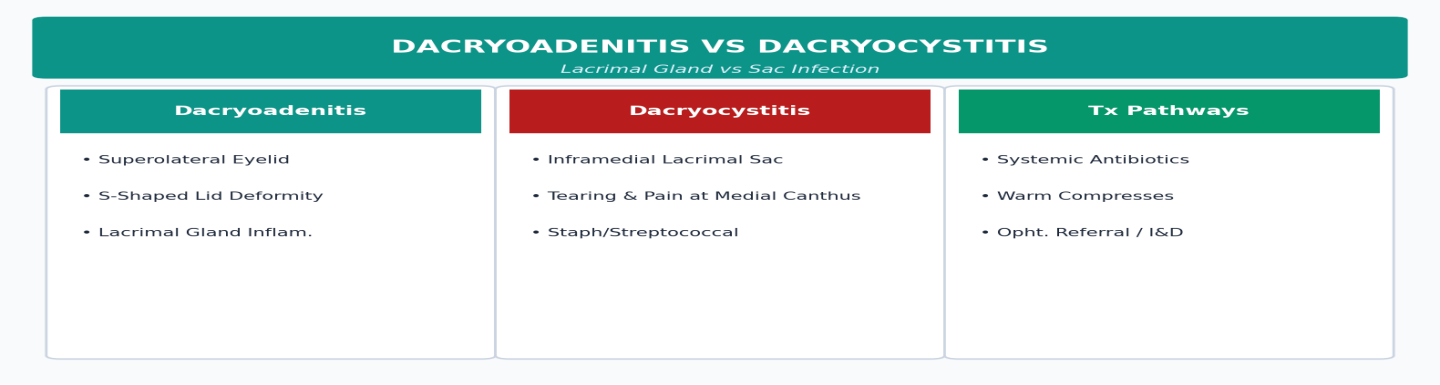
Dacryoadenitis: Erythema and swelling of lateral 1/3 of upper lid ('S-shaped' deformity). Dacryocystitis: Tender red nodule at medial canthus; purulent discharge can be expressed from punctum.

DIAGNOSTIC STUDIES:

Clinical diagnosis. CT orbit if orbital cellulitis or abscess formation is suspected.

MANAGEMENT & TREATMENT:

Systemic oral antibiotics (Augmentin or Cephalexin), warm compresses. Surgical incision and drainage or dacryocystorhinostomy (DCR) for dacryocystitis after acute infection resolves.



BOARD PEARL: Dacryoadenitis occurs in the SUPEROLATERAL lid (gland); Dacryocystitis occurs in the INFRAMEDIAL canthus (sac)!

PATHOPHYSIOLOGY & ETIOLOGY:

Periapical/Periodontal Abscess: Polymicrobial odontogenic infection (oral anaerobes, Streptococci). Ludwig's Angina: Rapidly spreading bilateral gangrenous cellulitis of the submandibular, sublingual, and submental spaces.

CLINICAL PRESENTATION:

Dental Abscess: Severe throbbing toothache, pain with chewing/percussion, localized facial swelling. Ludwig Angina: Fever, neck pain, dysphagia, drooling, trismus, and progressive 'woody' bilateral submandibular swelling.

PHYSICAL EXAMINATION:

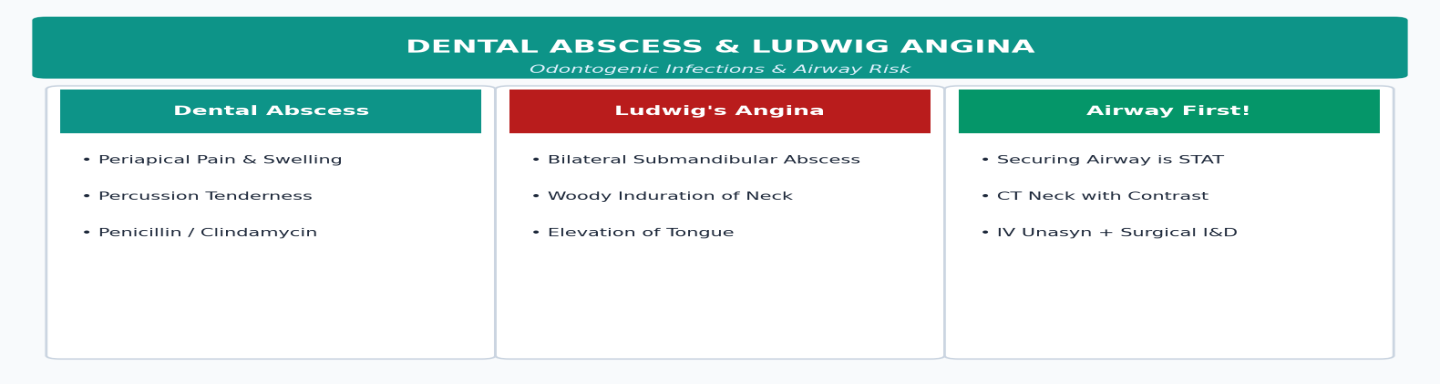
Ludwig Angina: Elevation and posterior displacement of the tongue threatening the airway, brawny induration of the anterior neck, floor of mouth erythema/edema.

DIAGNOSTIC STUDIES:

Dental Abscess: Dental X-rays or clinical. Ludwig Angina: Clinical emergency; CT neck with contrast AFTER securing airway.

MANAGEMENT & TREATMENT:

Dental Abscess: Penicillin VK, Amoxicillin, or Clindamycin; dental referral for drainage/root canal. Ludwig Angina: STAT airway management (awake fiberoptic intubation/tracheostomy), IV Unasyn or Clindamycin + Cefepime, emergency surgical I&D.;



BOARD PEARL: In suspected Ludwig's Angina, AIRWAY MANAGEMENT takes precedence over all imaging and diagnostic procedures!

PATHOPHYSIOLOGY & ETIOLOGY:

Severe, life-threatening cellulitis of the epiglottis and supraglottic structures. Historically caused by *Haemophilus influenzae* type b (Hib); now seen in immunized children and adults due to Group A Strep, *S. pneumo*, and *S. aureus*.

CLINICAL PRESENTATION:

Rapid onset high fever, severe sore throat, odynophagia, and the classic 3 Ds: Dysphagia, Drooling, and Distress. Patient assumes 'Tripod position' (leaning forward, neck extended).

PHYSICAL EXAMINATION:

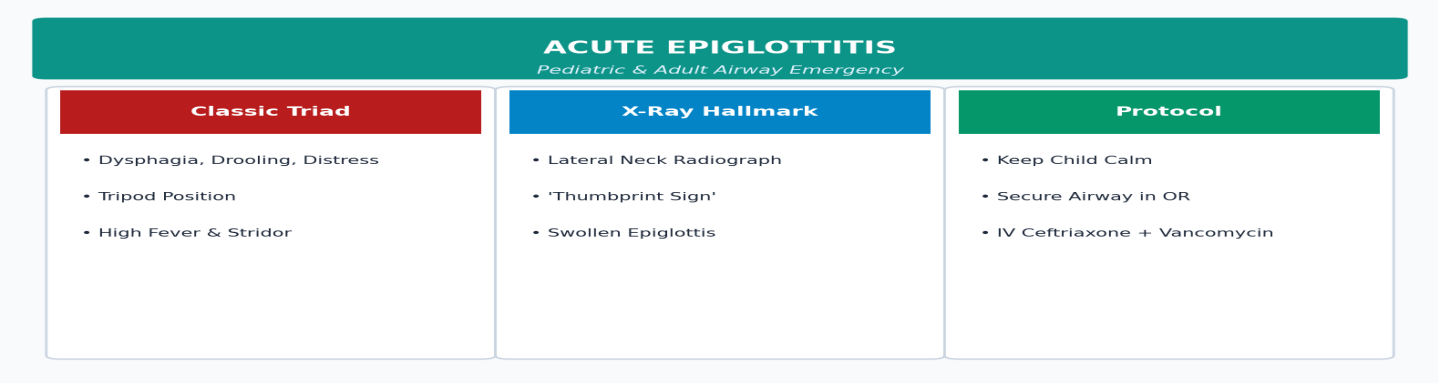
Inspiratory stridor, muffled 'hot potato' voice, anxiety. DO NOT examine pharynx with tongue depressor in pediatric patients as it may trigger fatal laryngospasm!

DIAGNOSTIC STUDIES:

Lateral Neck Radiograph demonstrates the classic 'Thumbprint sign' (swollen epiglottis). Definitive diagnosis via direct laryngoscopy in controlled OR setting.

MANAGEMENT & TREATMENT:

Keep patient calm (avoid IV lines or crying in children). Secure airway in the Operating Room with Anesthesiology/ENT. IV Ceftriaxone plus Vancomycin.



BOARD PEARL: Presence of drooling, stridor, and tripod positioning in a patient with a normal-appearing throat is Epiglottitis until proven otherwise!

PATHOPHYSIOLOGY & ETIOLOGY:

Nasal bleeding. Anterior Epistaxis (90%): Originates from Kiesselbach's Plexus (Anterior ethmoid, Sphenopalatine, Greater palatine, Superior labial arteries). Posterior Epistaxis (10%): Originates from Woodruff's Plexus (Sphenopalatine artery).

CLINICAL PRESENTATION:

Unilateral or bilateral nasal bleeding. Posterior epistaxis presents with profuse bleeding flowing down the posterior pharynx, unaffected by anterior pressure.

PHYSICAL EXAMINATION:

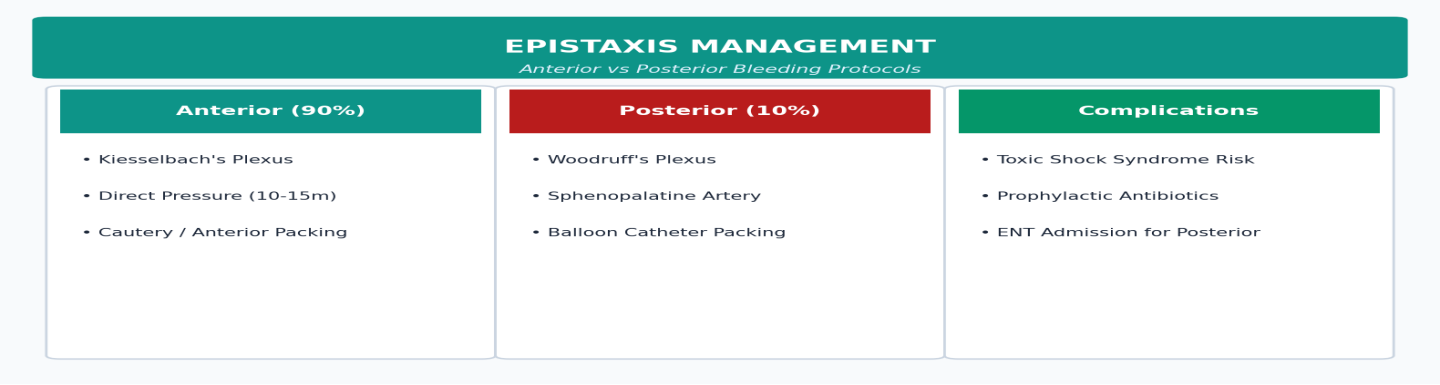
Examine with nasal speculum and headlight after blowing out clots and applying topical vasoconstrictor/anesthetic (Oxymetazoline/Lidocaine).

DIAGNOSTIC STUDIES:

Clinical evaluation. Hemoglobin/hematocrit, Coagulation panel (PT/INR) if on anticoagulants or severe bleeding.

MANAGEMENT & TREATMENT:

Anterior: Direct nasal compression (10-15 mins leaning forward) -> Chemical cautery with Silver Nitrate -> Anterior nasal packing (Rapid Rhino / Nasal Tampon). Posterior: Balloon catheter packing, ENT admission, prophylactic antibiotics (prevent Toxic Shock Syndrome).



BOARD PEARL: All patients with posterior nasal packing require hospital admission and continuous pulse oximetry due to risk of nasopulmonary reflex hypoventilation!

PATHOPHYSIOLOGY & ETIOLOGY:

Insertion or aspiration of foreign objects into ear canal, nasal cavity, or eye. Common in pediatric populations.

CLINICAL PRESENTATION:

Nasal FB: Unilateral foul-smelling purulent nasal discharge in a young child. Ear FB: Otalgia, fullness, decreased hearing, live insect sensation. Ocular FB: Pain, photophobia, tearing, foreign body sensation.

PHYSICAL EXAMINATION:

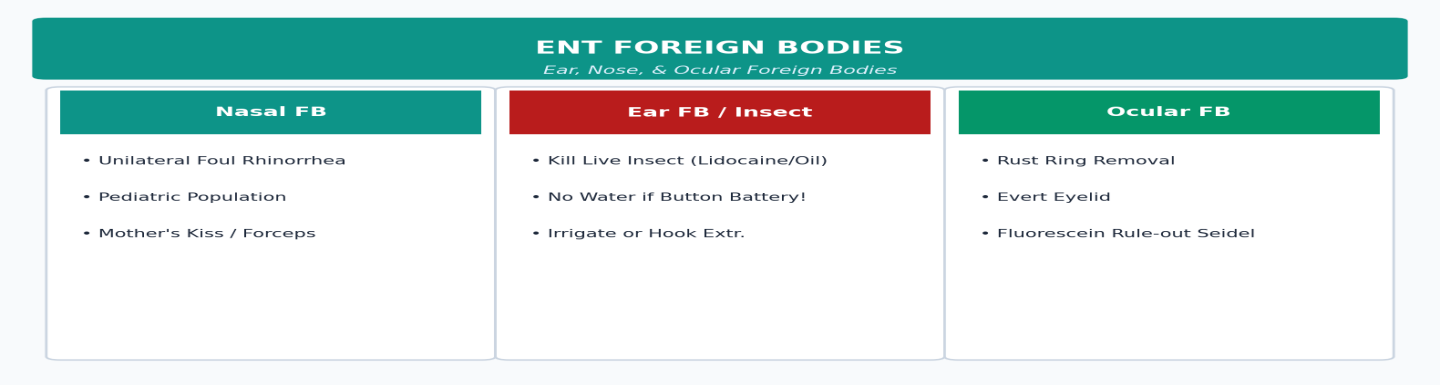
Nasal/Ear Speculum exam. Ocular: Slit lamp exam with fluorescein staining; evert eyelids to inspect palpebral conjunctiva.

DIAGNOSTIC STUDIES:

Clinical visualization. Plain X-rays for radiopaque objects or suspected button batteries.

MANAGEMENT & TREATMENT:

Nasal: 'Mother's kiss' technique, Katz extractor, or alligator forceps. Ear: Kill live insects with mineral oil or lidocaine before removal; NEVER irrigate organic matter (swells) or BUTTON BATTERIES (liquefactive necrosis!). Ocular: Irrigate or remove with cotton swab/needle; rust ring removal.



BOARD PEARL: Unilateral foul-smelling nasal discharge in a child is a NASAL FOREIGN BODY until proven otherwise! Button batteries in nose/ear are surgical emergencies!

PATHOPHYSIOLOGY & ETIOLOGY:

Ophthalmologic emergency caused by mechanical obstruction of the trabecular meshwork by the iris, preventing aqueous humor drainage and causing a rapid, severe spike in Intraocular Pressure (IOP >40-50 mmHg).

CLINICAL PRESENTATION:

Sudden onset severe unilateral eye pain, headache, nausea, vomiting, blurred vision, and seeing 'halos around lights'. Triggered by pupillary dilation (dark room, anticholinergic drugs).

PHYSICAL EXAMINATION:

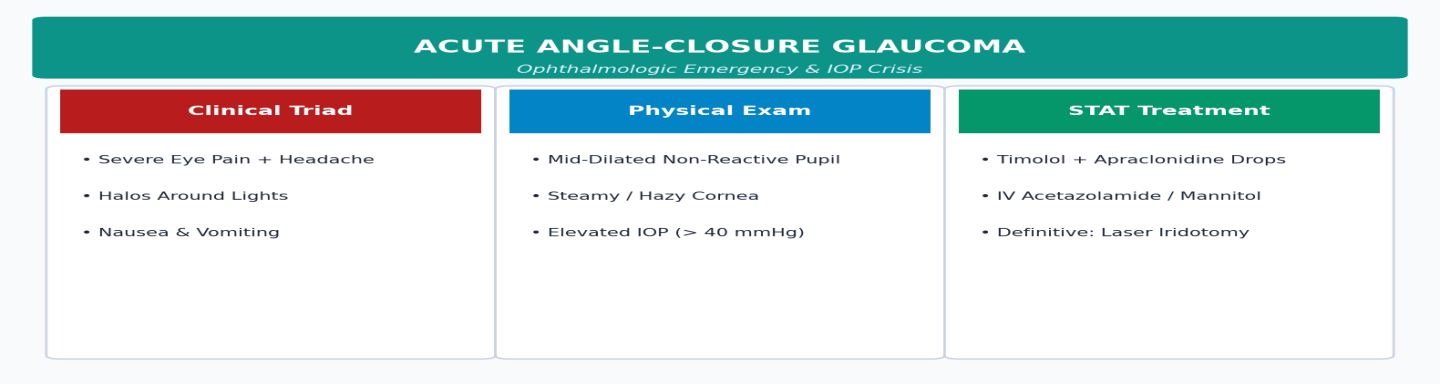
Mid-dilated, fixed, non-reactive pupil, steamy/hazy cornea, conjunctival injection (ciliary flush), and eye feels hard on palpation ('rock-hard globe').

DIAGNOSTIC STUDIES:

Tonometry demonstrates markedly elevated IOP (Normal = 10-21 mmHg; Angle-Closure = >40-50 mmHg).

MANAGEMENT & TREATMENT:

Immediate medical therapy to lower IOP: Topical Timolol (beta-blocker), Apraclonidine (alpha-2 agonist), Pilocarpine (miotic agent), and IV Acetazolamide or IV Mannitol. Definitive treatment: Laser Peripheral Iridotomy.



BOARD PEARL: Classic vignette: An elderly patient who develops severe eye pain, headache, and nausea after watching a movie in a dark theater!

PATHOPHYSIOLOGY & ETIOLOGY:

Accumulation of blood in the anterior chamber of the eye, usually caused by blunt ocular trauma disrupting the blood vessels of the ciliary body or iris.

CLINICAL PRESENTATION:

Eye pain, photophobia, somnolence (in young children), and decreased visual acuity following facial/ocular trauma.

PHYSICAL EXAMINATION:

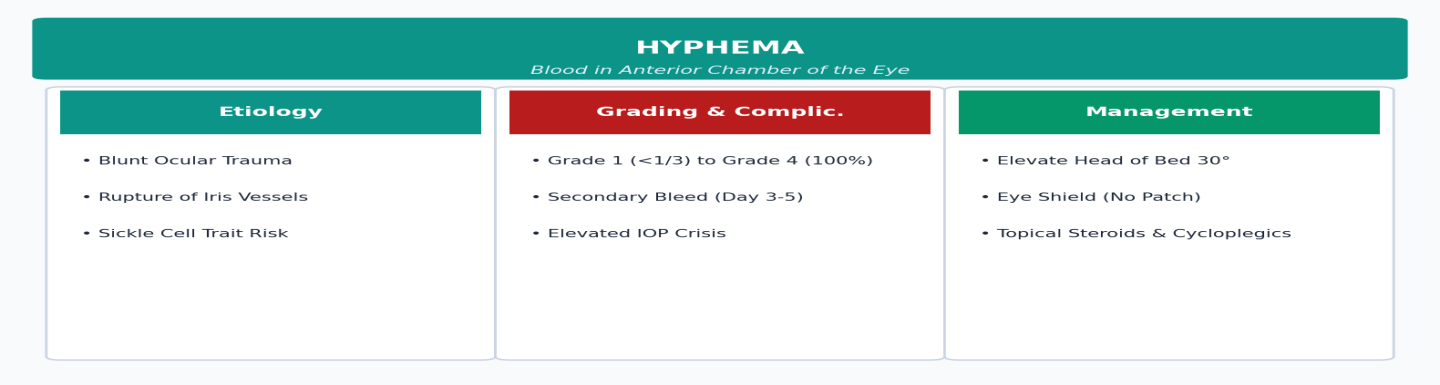
Visible layered blood level in the lower portion of the anterior chamber on penlight/slit-lamp exam. Graded 1 (<1/3 chamber) to 4 (100% '8-ball hyphema').

DIAGNOSTIC STUDIES:

Clinical examination. Orbital CT if associated facial/orbital floor fractures suspected. Screen for Sickle Cell trait/disease.

MANAGEMENT & TREATMENT:

Elevate head of bed 30° (promotes blood settling), place rigid eye shield (NO patch!), topical cycloplegics (Atropine) and topical steroids. Avoid NSAIDs/Aspirin (worsens bleeding). Monitor for secondary re-bleeding (Days 3-5).



BOARD PEARL: Patients with Sickle Cell Trait or Disease and hyphema are at extreme risk for secondary glaucoma and optic nerve atrophy even with mild IOP elevations!

PATHOPHYSIOLOGY & ETIOLOGY:

Inflammatory disorders of the inner ear, often post-viral. Vestibular Neuritis: Inflammation of the vestibular branch of CN VIII. Labyrinthitis: Inflammation of BOTH vestibular and cochlear branches of CN VIII.

CLINICAL PRESENTATION:

Sudden onset continuous, severe, incapacitating vertigo, nausea, vomiting, gait instability, and spontaneous horizontal nystagmus. KEY DIFFERENCE: Labyrinthitis has UNILATERAL HEARING LOSS and tinnitus; Vestibular Neuritis has NORMAL HEARING.

PHYSICAL EXAMINATION:

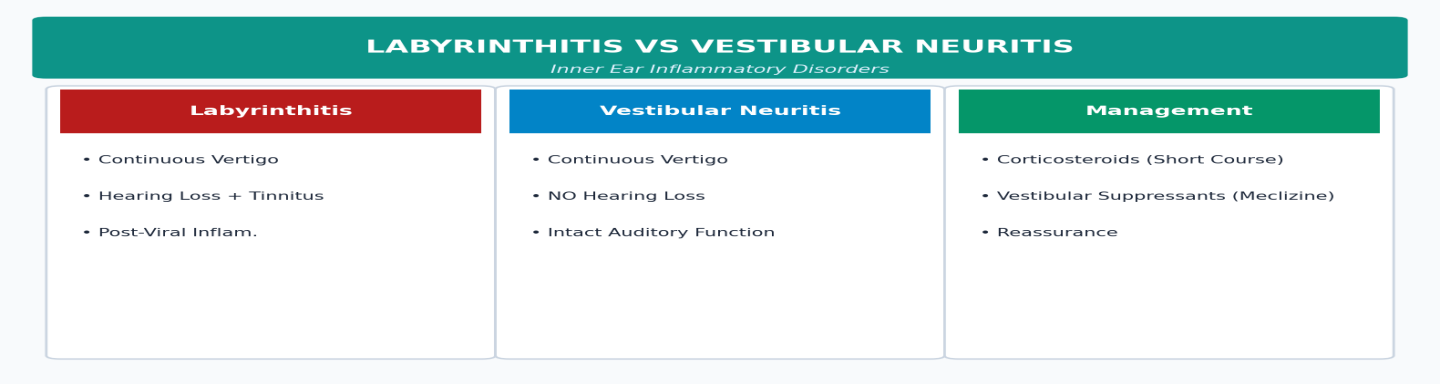
Spontaneous horizontal-torsional nystagmus (suppressed by visual fixation). HINTS exam shows abnormal Head Impulse test (saccade toward affected side). No focal neurological deficits.

DIAGNOSTIC STUDIES:

Clinical diagnosis. MRI Brain if central cause (e.g., cerebellar stroke) cannot be excluded.

MANAGEMENT & TREATMENT:

Short course of corticosteroids (Prednisone), vestibular suppressants for acute symptoms only (Meclizine, Diazepam, Ondansetron for 24-48 hours). Vestibular rehabilitation.



BOARD PEARL: Sudden continuous vertigo WITH hearing loss = Labyrinthitis. Sudden continuous vertigo WITHOUT hearing loss = Vestibular Neuritis!

PATHOPHYSIOLOGY & ETIOLOGY:

Leading cause of permanent central vision loss in elderly patients. Dry (Atrophic) AMD (90%): Degeneration of retinal pigment epithelium with Drusen deposits. Wet (Neovascular/Exudative) AMD (10%): Choroidal neovascularization leaking fluid/blood under macula.

CLINICAL PRESENTATION:

Gradual (Dry) or Rapid (Wet) painless loss of CENTRAL vision. Metamorphopsia (distortion of straight lines appearing wavy). Peripheral vision remains intact.

PHYSICAL EXAMINATION:

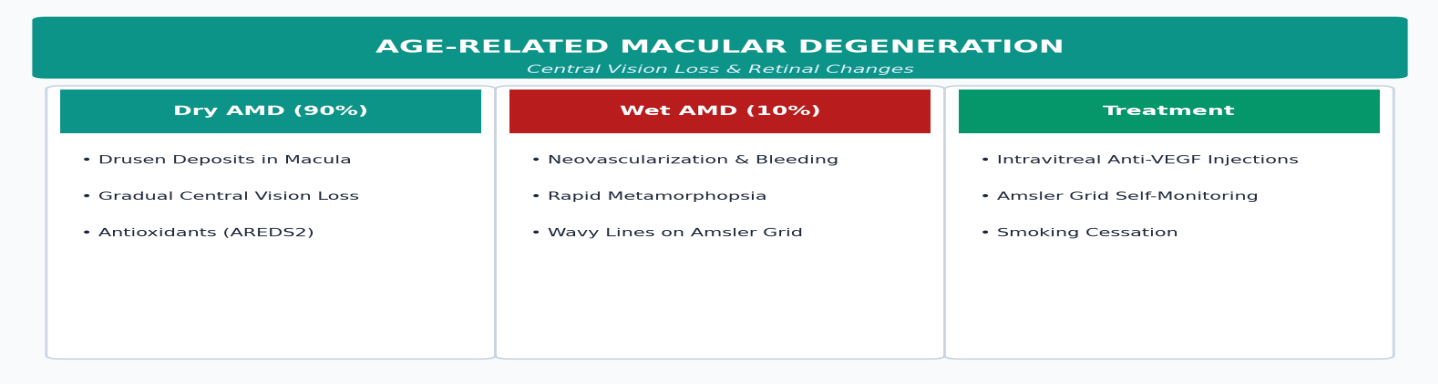
Fundoscopy: Dry AMD shows Drusen (yellow extracellular deposits in macula) and retinal pigment epithelial atrophy. Wet AMD shows subretinal fluid, hemorrhages, or gray-green choroidal neovascular membrane.

DIAGNOSTIC STUDIES:

Amsler Grid testing (wavy lines). Optical Coherence Tomography (OCT) and Fluorescein Angiography for wet AMD.

MANAGEMENT & TREATMENT:

Dry AMD: Oral antioxidant vitamins + zinc (AREDS2 formulation), smoking cessation. Wet AMD: Intravitreal Anti-VEGF Injections (Ranibizumab, Aflibercept).



BOARD PEARL: Metamorphopsia (straight lines appearing curved or wavy on an Amsler grid) is the key early symptom of Wet Macular Degeneration!

PATHOPHYSIOLOGY & ETIOLOGY:

Suppurative infection and inflammation of the mastoid air cells, occurring as a direct complication of untreated or inadequately treated Acute Otitis Media. Pathogens: *S. pneumoniae*, *S. pyogenes*, *P. aeruginosa*.

CLINICAL PRESENTATION:

Postauricular pain, erythema, swelling, fever, otorrhea, and forward displacement/proptosis of the pinna (auricle).

PHYSICAL EXAMINATION:

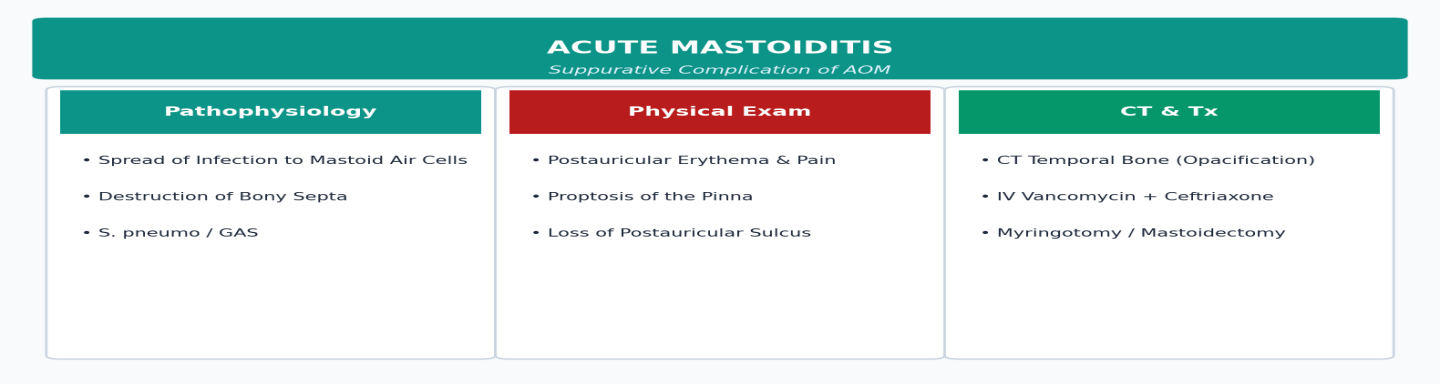
Postauricular tenderness, fluctuations, erythema, loss of postauricular crease, and abnormal TM findings consistent with AOM.

DIAGNOSTIC STUDIES:

CT Temporal Bone with Contrast: Demonstrates opacification of mastoid air cells with destruction of bony septa (coalescent mastoiditis).

MANAGEMENT & TREATMENT:

Emergency ENT consultation. IV broad-spectrum antibiotics (Vancomycin + Ceftriaxone or Cefepime). Myringotomy with tympanostomy tube placement; Mastoidectomy for refractory cases or subperiosteal abscess.



BOARD PEARL: Forward displacement of the pinna (auricle) with postauricular tenderness is the classic physical exam finding of Acute Mastoiditis!

PATHOPHYSIOLOGY & ETIOLOGY:

Demyelinating inflammation of the optic nerve (CN II). Highly associated with Multiple Sclerosis (MS) — Optic Neuritis is the presenting symptom in 20% of MS patients.

CLINICAL PRESENTATION:

Acute, unilateral loss of vision (developing over hours to days), eye pain aggravated by eye movements, and marked loss of color vision (red desaturation).

PHYSICAL EXAMINATION:

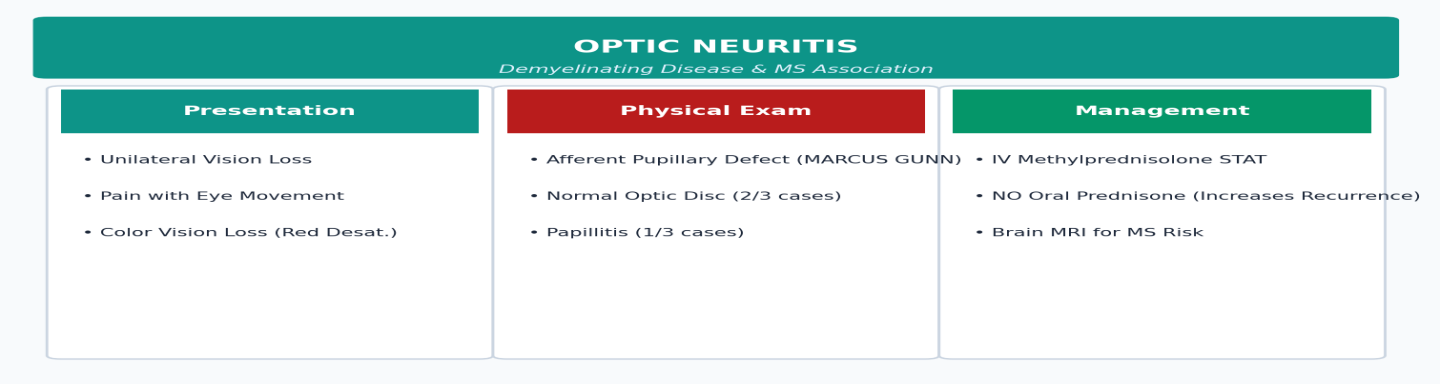
Relative Afferent Pupillary Defect (RAPD / Marcus Gunn Pupil). Fundoscopy: Normal optic disc in 2/3 of cases (Retrobulbar Neuritis: 'patient sees nothing, doctor sees nothing'); Papillitis (swollen disc) in 1/3.

DIAGNOSTIC STUDIES:

Clinical exam. MRI of Brain and Orbits with gadolinium: Confirms optic nerve enhancement and detects brain demyelinating lesions.

MANAGEMENT & TREATMENT:

STAT IV Methylprednisolone (1000 mg/day for 3 days) followed by oral prednisone taper. Oral prednisone ALONE is contraindicated as it increases recurrence rates!



BOARD PEARL: Eye pain that worsens with ocular movements combined with red color desaturation in a young female is classic for Optic Neuritis!

PATHOPHYSIOLOGY & ETIOLOGY:

Preseptal (Periorbital) Cellulitis: Infection anterior to the orbital septum (skin/eyelid trauma). Orbital Cellulitis: Infection POSTERIOR to orbital septum, usually secondary to ethmoid sinusitis (*S. aureus*, *S. pneumo*).

CLINICAL PRESENTATION:

Preseptal: Eyelid erythema and edema, no vision changes. Orbital: Eyelid swelling PLUS proptosis, ophthalmoplegia (painful/restricted eye movements), and decreased visual acuity.

PHYSICAL EXAMINATION:

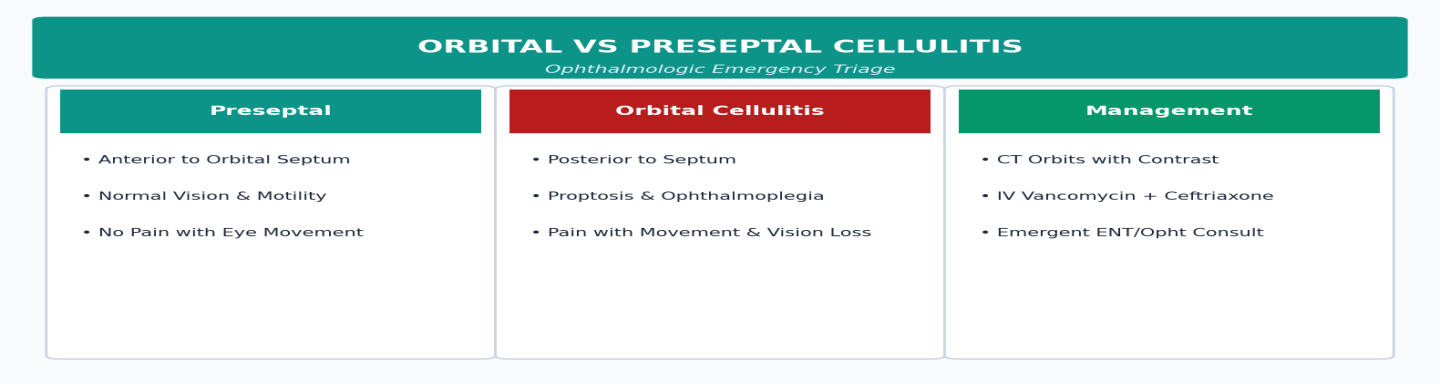
Orbital Cellulitis presents with Proptosis, Pain with Extraocular Movements, Fever, and Decreased Vision. Preseptal lacks these 4 features!

DIAGNOSTIC STUDIES:

High-resolution CT Scan of Orbits and Sinuses with IV Contrast: Differentiates preseptal from orbital cellulitis and detects subperiosteal abscess.

MANAGEMENT & TREATMENT:

Preseptal: Outpatient oral antibiotics (Augmentin or Cefpodoxime). Orbital Cellulitis: Inpatient IV Vancomycin PLUS Ceftriaxone/Unasyn, emergency ENT/Ophthalmology consult for surgical drainage if abscess present.



BOARD PEARL: Pain with eye movement and proptosis are the two key physical exam findings that distinguish Orbital Cellulitis from Preseptal Cellulitis!

PATHOPHYSIOLOGY & ETIOLOGY:

Acute Otitis Externa (Swimmer's Ear): Infection of external auditory canal (*Pseudomonas aeruginosa*, *S. aureus*). Malignant (Necrotizing) Otitis Externa: Invasive osteomyelitis of skull base in elderly diabetics/immunocompromised (*P. aeruginosa*).

CLINICAL PRESENTATION:

Uncomplicated: Severe ear pain, itching, otorrhea, and pain with tragal movement. Malignant: Severe deep otalgia out of proportion, foul purulent discharge, granulations in ear canal, and facial nerve palsy (CN VII).

PHYSICAL EXAMINATION:

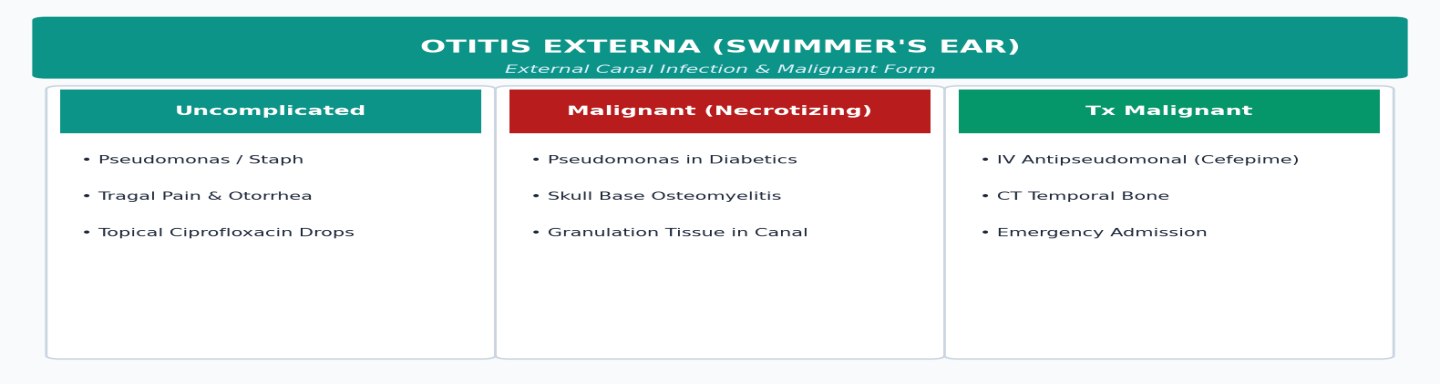
Uncomplicated: Swollen, edematous ear canal with debris; exquisite pain on pulling pinna or pushing tragus. Malignant: Granulation tissue at junction of cartilaginous and bony canal.

DIAGNOSTIC STUDIES:

Clinical. Malignant OE: CT Temporal bone (bone erosion), elevated ESR/CRP.

MANAGEMENT & TREATMENT:

Uncomplicated: Topical antibiotic/steroid drops (Ciprofloxacin/Dexamethasone). Malignant OE: Admission, STAT IV antipseudomonal antibiotics (Cefepime, Piperacillin-Tazobactam, or Ciprofloxacin).



BOARD PEARL: Presence of granulation tissue in the external auditory canal of an elderly diabetic patient with severe otalgia indicates Malignant Otitis Externa!

PATHOPHYSIOLOGY & ETIOLOGY:

Bilateral optic disc swelling secondary to Increased Intracranial Pressure (ICP). Causes: Brain tumor, intracranial hemorrhage, meningitis, venous sinus thrombosis, Idiopathic Intracranial Hypertension (Pseudotumor Cerebri).

CLINICAL PRESENTATION:

Headache (worse in morning/coughing), transient visual obscurations (brief graying out of vision with posture changes), nausea, vomiting, and diplopia (CN VI palsy).

PHYSICAL EXAMINATION:

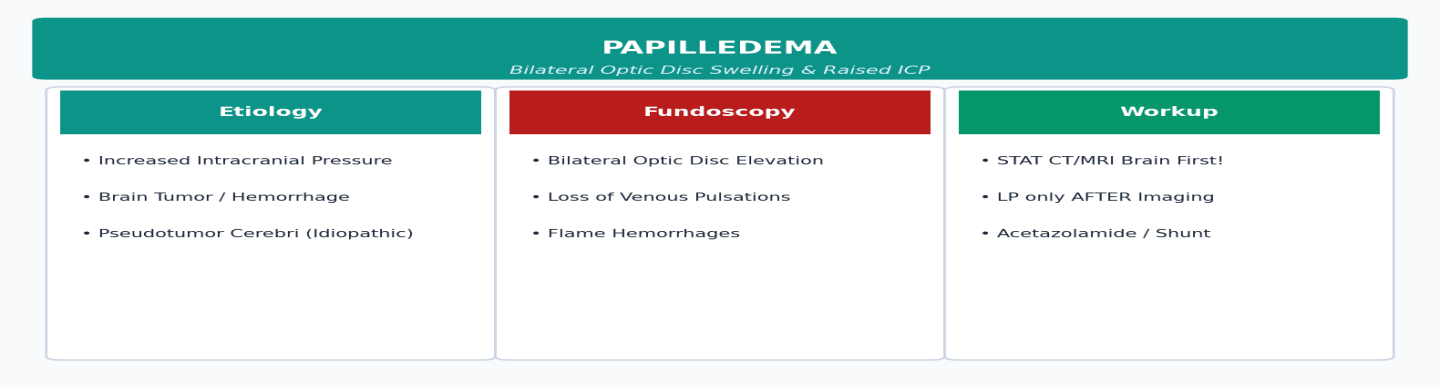
Fundoscopy: Bilateral optic disc elevation, blurred disc margins, loss of spontaneous venous pulsations, flame-shaped hemorrhages, and engorged retinal veins.

DIAGNOSTIC STUDIES:

Emergency Neuroimaging (CT or MRI Brain) MUST be performed FIRST to rule out mass lesion/herniation before Lumbar Puncture!

MANAGEMENT & TREATMENT:

Treat underlying cause of elevated ICP. Pseudotumor Cerebri: Weight loss, Acetazolamide, serial LPs, or VP shunt.



BOARD PEARL: Always perform neuroimaging (CT/MRI) BEFORE Lumbar Puncture in a patient with papilledema to prevent fatal brain herniation!

PATHOPHYSIOLOGY & ETIOLOGY:

Collection of pus in the space between the tonsillar capsule and superior pharyngeal constrictor muscle. Most common deep neck space infection. Polymicrobial (Group A Strep, *S. aureus*, anaerobes).

CLINICAL PRESENTATION:

Severe unilateral sore throat, fever, odynophagia, trismus (difficulty opening mouth), muffled 'hot potato' voice, drooling, and ipsilateral ear pain.

PHYSICAL EXAMINATION:

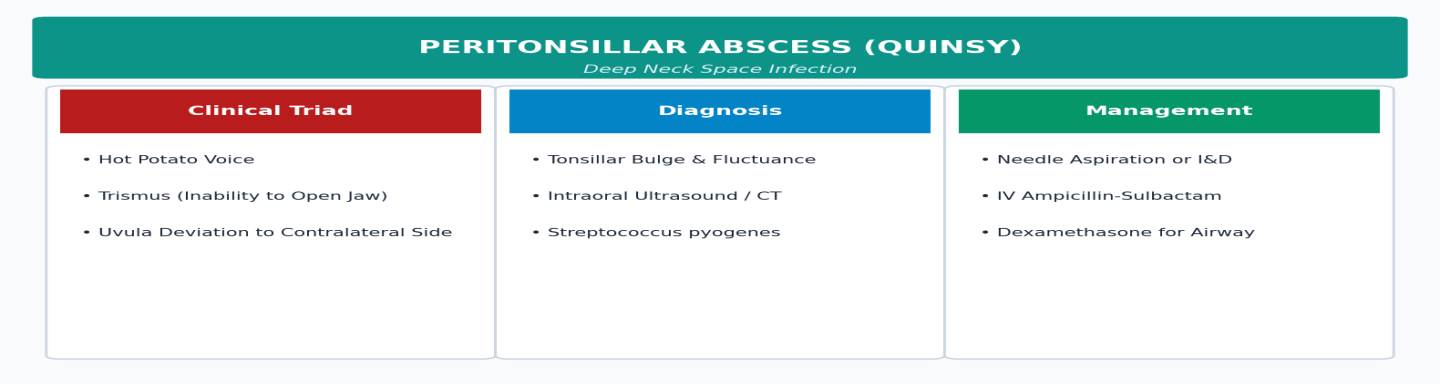
Unilateral tonsillar swelling with medial displacement of the tonsil, bulging soft palate, and deviation of the uvula to the CONTRALATERAL side.

DIAGNOSTIC STUDIES:

Clinical diagnosis. Intraoral Ultrasound or CT Neck with contrast for confirmation.

MANAGEMENT & TREATMENT:

Needle aspiration or Incision & Drainage (I&D;). IV Antibiotics (Ampicillin-Sulbactam or Clindamycin) and IV Dexamethasone to reduce airway edema.



BOARD PEARL: Trismus, a 'hot potato' voice, and uvular deviation away from the affected side are the diagnostic hallmark triad of Peritonsillar Abscess!

PATHOPHYSIOLOGY & ETIOLOGY:

Separation of the neurosensory retina from the underlying retinal pigment epithelium. Rhegmatogenous (most common; retinal tear allowing vitreous fluid under retina; risk factors: myopia, cataract surgery, trauma).

CLINICAL PRESENTATION:

Painless vision loss preceded by Photopsia (flashing lights), sudden increase in Floaters ('cobwebs'), followed by a 'shadow' or 'curtain' dropping over the field of vision.

PHYSICAL EXAMINATION:

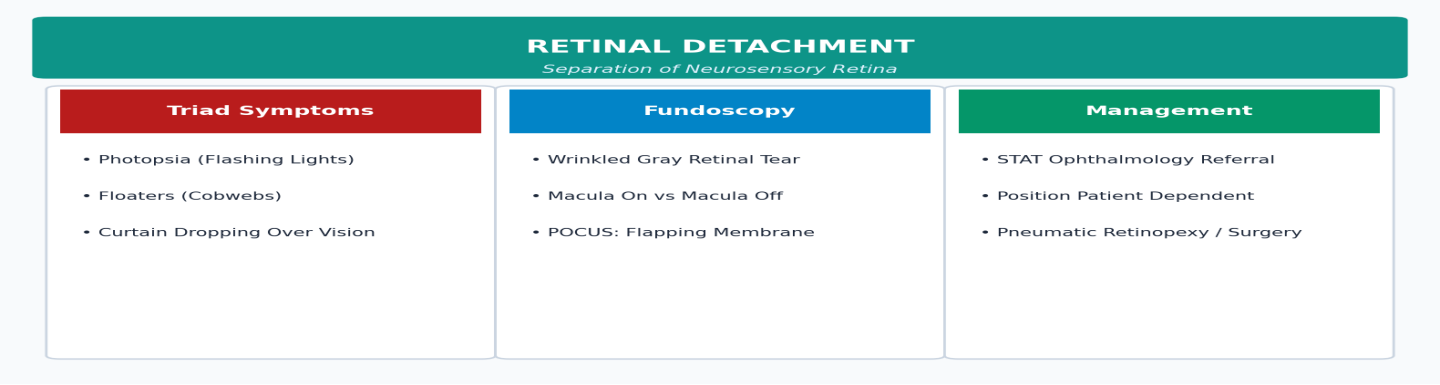
Fundoscopy: Elevated, wrinkled, gray, opalescent retina with dark folded blood vessels. Ocular POCUS shows hyperechoic membrane flapping in vitreous chamber.

DIAGNOSTIC STUDIES:

Dilated Fundoscopic Exam by Ophthalmology. Bedside Ocular Ultrasound.

MANAGEMENT & TREATMENT:

Emergency Ophthalmology Consult. Keep patient supine with head turned toward side of detachment. Surgical repair (Pneumatic retinopexy, scleral buckle, or vitrectomy).



BOARD PEARL: Classic history: 'A dark curtain came down over my eye after seeing bright flashes of light and floating spots.'

PATHOPHYSIOLOGY & ETIOLOGY:

Central Retinal Artery Occlusion (CRAO): Embolic/thrombotic occlusion of CRA (ocular stroke; carotid plaque/AFib). Central Retinal Vein Occlusion (CRVO): Thrombosis of CRV (associated with HTN, glaucoma, hypercoagulability).

CLINICAL PRESENTATION:

CRAO: Sudden, painless, severe monocular vision loss. CRVO: Sudden or subacute monocular vision loss.

PHYSICAL EXAMINATION:

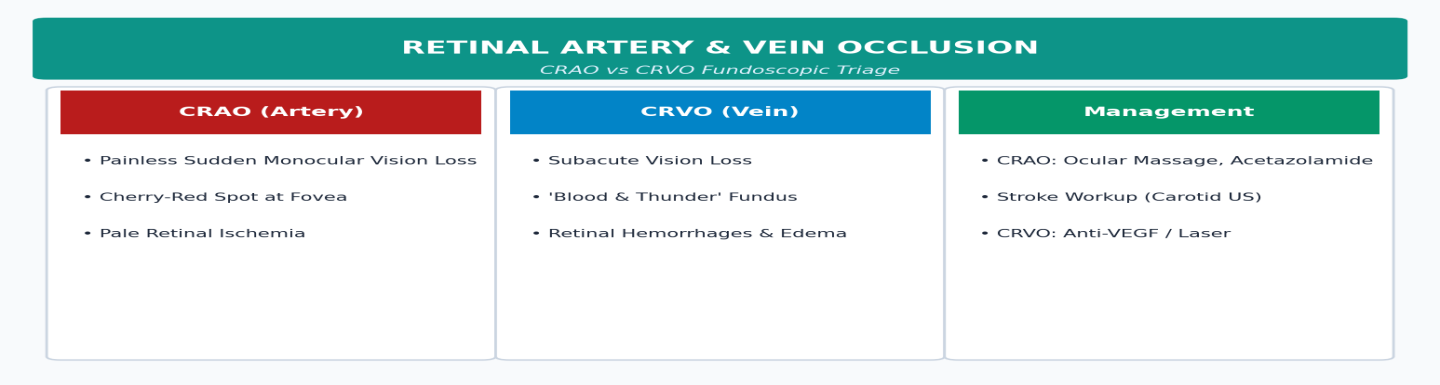
CRAO Fundoscopy: Pale, edematous retina with a classic 'Cherry-red spot' at the fovea, boxcarring of retinal vessels. CRVO Fundoscopy: Extensively hemorrhaged retina ('Blood and Thunder' fundus), optic disc edema, cotton wool spots.

DIAGNOSTIC STUDIES:

Fundoscopic exam. CRAO requires emergent stroke workup (Carotid duplex, Echo, EKG).

MANAGEMENT & TREATMENT:

CRAO: Emergent Ophthalmology consult; ocular massage, IV Acetazolamide/Mannitol, carbogen inhalation (aim to restore flow within 90 mins). CRVO: Anti-VEGF intravitreal injections, laser photocoagulation.



BOARD PEARL: CRAO = 'Cherry-Red Spot' on pale retina. CRVO = 'Blood and Thunder' extensive retinal hemorrhages!

PATHOPHYSIOLOGY & ETIOLOGY:

Blunt trauma to the auricle (wrestling, boxing) causes subperichondrial hematoma formation, shearing the perichondrium from the avascular cartilage and causing cartilage necrosis.

CLINICAL PRESENTATION:

Tender, edematous, purple, tense swelling over the anterior aspect of the pinna following blunt ear trauma.

PHYSICAL EXAMINATION:

Fluctuant, erythematous or violaceous mass obscuring normal anatomical landmarks of the auricle.

DIAGNOSTIC STUDIES:

Clinical evaluation.

MANAGEMENT & TREATMENT:

Immediate Incision & Drainage (or needle aspiration for small hematomas <24 hrs) followed by Placement of a Compression Bolster Dressing (prevents re-accumulation). Oral antibiotics covering *Pseudomonas* (Ciprofloxacin).



BOARD PEARL: Inadequately drained auricular hematomas lead to aseptic cartilage necrosis and permanent 'Cauliflower Ear' (wrestler's ear) deformity!

PATHOPHYSIOLOGY & ETIOLOGY:

Rupture of the tympanic membrane caused by severe AOM infection, sudden pressure change (barotrauma, slap to ear), or direct penetrating trauma (Q-tips, bobby pins).

CLINICAL PRESENTATION:

Sudden onset severe ear pain followed by sudden RELIEF of pain accompanied by bloody or purulent otorrhea, conductive hearing loss, and tinnitus.

PHYSICAL EXAMINATION:

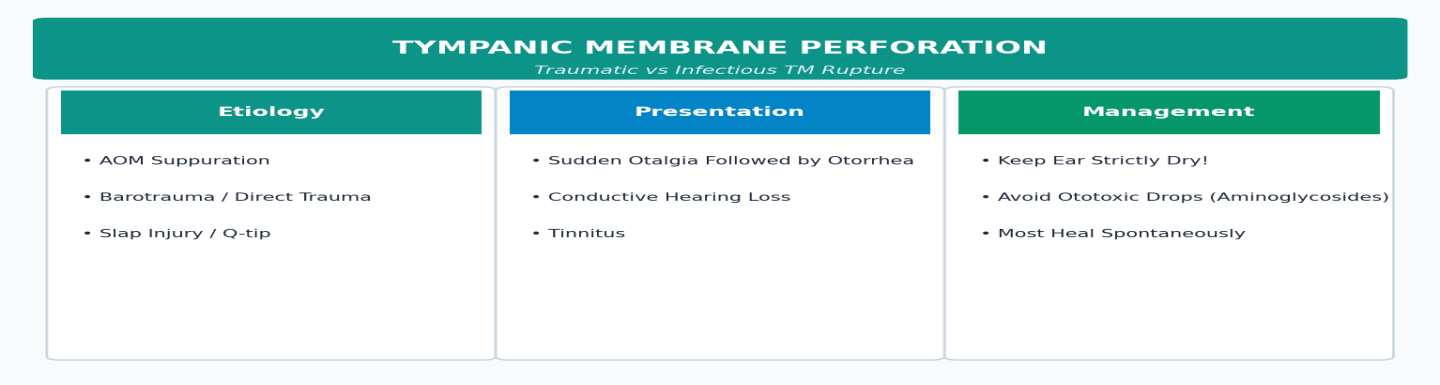
Otoscopy reveals a visible tear/hole in the TM with blood or purulent fluid in the canal.

DIAGNOSTIC STUDIES:

Clinical examination with otoscopy. Audiometry if persistent hearing loss.

MANAGEMENT & TREATMENT:

Keep the ear strictly DRY (no swimming/showering without earplugs!). Avoid ototoxic topical drops (Aminoglycosides). Most perforations heal spontaneously within 2-3 weeks. ENT follow-up.



BOARD PEARL: Sudden cessation of severe otalgia accompanied by purulent otorrhea in a patient with AOM indicates acute TM perforation!

PATHOPHYSIOLOGY & ETIOLOGY:

Perception of movement when none exists. Peripheral Vertigo (85%): BPPV, Meniere, Labyrinthitis, Vestibular Neuritis. Central Vertigo (15%): Cerebellar stroke, Wallenberg syndrome, MS, Brainstem tumor.

CLINICAL PRESENTATION:

Peripheral: Sudden onset, severe, episodic, horizontal/rotary fatigable nystagmus, inhibited by visual fixation. Central: Gradual/sudden onset, persistent, vertical non-fatigable nystagmus, NOT inhibited by fixation, associated with 5 Ds (Diplopia, Dysarthria, Dysphagia, Dysmetria, Dizziness).

PHYSICAL EXAMINATION:

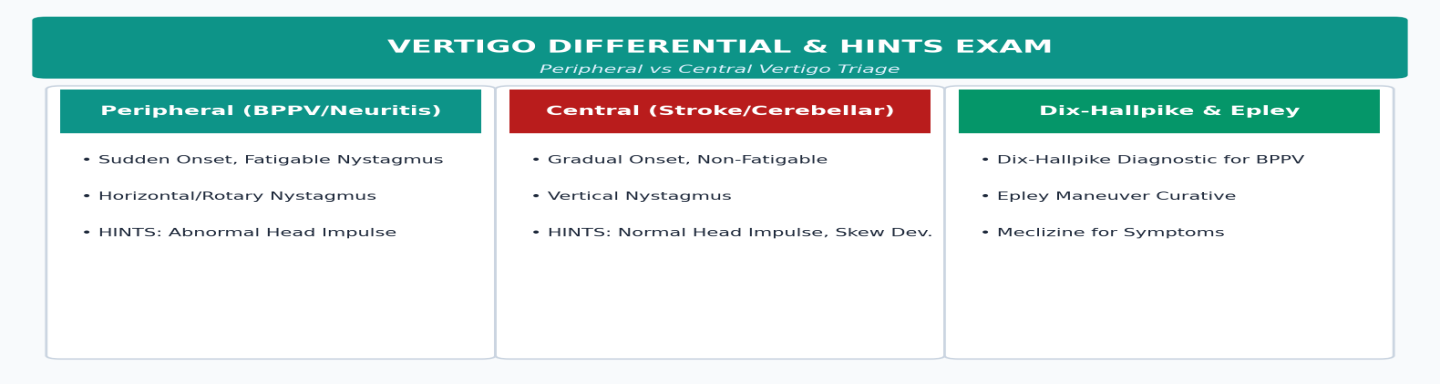
HINTS Exam (Head Impulse, Nystagmus, Test of Skew): Central stroke is suggested by (1) Normal Head Impulse Test; (2) Direction-changing/Vertical Nystagmus; or (3) Skew Deviation.

DIAGNOSTIC STUDIES:

Dix-Hallpike Maneuver diagnostic for BPPV (geotropic fatigable nystagmus). STAT Brain MRI for suspected central vertigo.

MANAGEMENT & TREATMENT:

BPPV: Epley Maneuver (repositioning canaliths) is curative. Symptomatic: Antihistamines (Meclizine), Benzodiazepines, Antiemetics.



BOARD PEARL: A NORMAL Head Impulse Test in a patient with continuous acute vestibular syndrome strongly points to a CENTRAL CEREBELLAR STROKE!

PATHOPHYSIOLOGY & ETIOLOGY:

Sudden loss of visual acuity. Categorized into Painless vs Painful Monocular Vision Loss.

CLINICAL PRESENTATION:

Painless Monocular Vision Loss: Central Retinal Artery Occlusion (CRAO), Central Retinal Vein Occlusion (CRVO), Retinal Detachment, Ischemic Optic Neuropathy (GCA). Painful Monocular Vision Loss: Acute Angle-Closure Glaucoma, Optic Neuritis, Corneal Ulcer, Endophthalmitis.

PHYSICAL EXAMINATION:

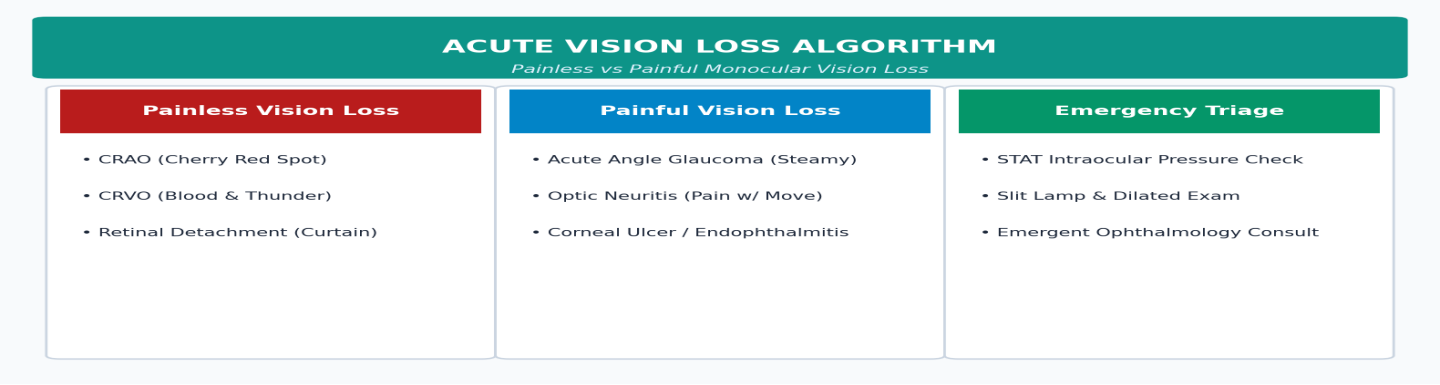
Assess Visual Acuity, Pupillary Reflexes (RAPD), Intraocular Pressure (Tonometry), Slit-Lamp Exam, and Fundoscopy.

DIAGNOSTIC STUDIES:

Emergent bedside Triage: Check IOP immediately (Glaucoma), check Temporal Artery ESR/CRP (Giant Cell Arteritis in elderly), Fundoscopy.

MANAGEMENT & TREATMENT:

Emergent Ophthalmology Consultation. Immediate intervention based on etiology (IOP lowering for Glaucoma; High-dose IV Steroids for Giant Cell Arteritis / Optic Neuritis; Retinal surgery for Detachment).



BOARD PEARL: In any elderly patient (>50) presenting with sudden painless vision loss, immediately order ESR and CRP to rule out Giant Cell (Temporal) Arteritis!